

CHENRUI LIU

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EDUCATION

University of California, San Diego

M.S. in Computer Science, GPA: 4.0/4.0

San Diego, CA

Sept 2025 – Mar/Jun 2027 (Expected)

Southeast University

B.Eng. in Computer Science and Technology, GPA: 3.81/4.0

Nanjing, China

Sept 2021 – June 2025

- Exchange Program at Nanyang Technological University, Singapore (Jan 2025 – May 2025)
- Exchange Program at University of California, San Diego (Mar 2024 – June 2024)

EXPERIENCE

China Telecom Hongxin Information Technology Co., Ltd.

July 2024 – Jan 2025

Software Development Intern

Nanjing, China

- Developed a questionnaire-driven lawsuit document generation module using Spring Boot and MyBatis-Plus.
- Implemented database-driven branching logic that dynamically selected subsequent questions based on user responses, enabling configurable decision-tree workflows.
- Designed multi-select branch processing by caching user-specific selections and sequentially resolving dependent question paths before document generation.
- Integrated Word template rendering, PDF conversion, and MinIO object storage, mapping questionnaire responses to template placeholders and enabling QR-code-based document downloads.

PROJECTS

Source code, demos, and additional projects are available at my profile 🌐 llcr6.github.io.

Paper Research Agent | Python, JavaScript, OpenAI Agents SDK, SQLite, Ollama

May 2026 – June 2026

- Built a local paper research chat app that helps users search arXiv, select papers, generate bibliography entries, and download PDFs through tool-calling agent workflows
- Implemented a Python backend with OpenAI-compatible and Ollama model support, persistent SQLite chat sessions, local state management, and reliable file handling
- Designed and developed a browser-based front end with chat history navigation, model settings panels, progress status updates, and responsive research conversation views
- Created an evaluation harness with multi-scenario tests and packaged the application as a macOS DMG for distribution;

LLM-Based Database Schema Linking | Python, PyTorch, SFT, PEFT

May 2026 – June 2026

- Built a schema-linking pipeline that predicts relevant database tables and columns from natural-language questions by prompting an instruction-tuned LLM with database schema context
- Fine-tuned Qwen2.5-1.5B-Instruct with LoRA adapters and compared model families, learning rates, epoch counts, LoRA ranks, and target modules
- Implemented deterministic inference and schema-aware post-processing to remove hallucinated identifiers, canonicalize casing, and reach a 0.4993 overall schema-linking score on validation;

RAG-Powered Documentation Q&A with Hybrid Retrieval | Python, RAG

Apr 2026 – May 2026

- Built an end-to-end RAG system for RapidFire AI documentation Q&A, generating grounded answers with source citations based on retrieved documentation chunks
- Implemented hybrid retrieval with BM25 and dense embeddings, fused results using Reciprocal Rank Fusion, and refined rankings with a cross-encoder reranker
- Conducted controlled experiments on chunk size, reranker threshold, and reranker models, achieving a best retrieval score of 0.7699 with MiniLM-L12-v2
- Created validation Q&A pairs and evaluation scripts for retrieval F1, precision, recall, and LLM-judge answer quality;

Memory-Efficient Micro-UAV Autonomous Navigation | C, FreeRTOS

Nov 2022 – Mar 2024

- Worked on an embedded autonomous navigation system for resource-constrained micro-UAVs using C and FreeRTOS, combining LiDAR-based 3D mapping with RRT* path planning.
- Designed a memory-efficient octree with a custom linear memory pool, allocating child nodes in contiguous groups of eight to reduce dynamic memory allocation overhead.
- Compressed each octree node into a 16-bit representation using C bit fields to encode child-unit indices, occupancy states, and leaf status, enabling efficient mapping under limited memory.
- Contributed to a granted Chinese invention patent for the system (CN116499464B).

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, SQL, HTML/CSS, TypeScript, LaTeX, MATLAB

Developer Tools: Git, Vim, VS Code, PyCharm, IntelliJ, CLion, Navicat, PostgreSQL, MySQL, SQLite, Codex, Claude Code

Frameworks & Libraries: Spring Boot, Flask, PyTorch, Pandas, Numpy, Matplotlib, Seaborn, Scikit-Learn, NLTK